Data Base Lab
Lab Task 8

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Section 4A

Question 01

```
MariaDB [lab_10_task]> USE lab_10_task;
Database changed
MariaDB [lab_10_task]> CREATE VIEW active_models AS
    -> SELECT m.model_name, p.provider_name, m.base_price
    -> FROM models m
    -> JOIN providers p ON m.provider_id = p.provider_id
    -> WHERE m.is_active = TRUE;
Query OK, 0 rows affected (0.020 sec)
```

Question 02

```
MariaDB [lab_10_task]> CREATE VIEW usage_summary AS
    -> SELECT m.model_name, COUNT(l.log_id) AS total_requests, SUM(l.cost) AS total_cost
    -> FROM models m
    -> LEFT JOIN usage_logs l ON m.model_id = l.model_id
    -> GROUP BY m.model_id, m.model_name;
Query OK, 0 rows affected (0.008 sec)

MariaDB [lab_10_task]>
```

Question 03

```
MariaDB [lab_10_task]> SELECT model_name, total_cost
    -> FROM usage_summary
    -> WHERE total_cost > (SELECT AVG(total_cost) FROM usage_summary);
+---------------+------------+
| model_name    | total_cost |
+---------------+------------+
| ChatGPT-4     |     0.0350 |
| Claude 3 Opus |     0.0300 |
| Gemini Pro    |     0.0250 |
+---------------+------------+
3 rows in set (0.004 sec)

MariaDB [lab_10_task]>
```

Question 04

```
MariaDB [lab_10_task]> SELECT CONCAT(user_name, ' used ', m.model_name) AS usage_statement
    -> FROM usage_logs l
    -> JOIN models m ON l.model_id = m.model_id;
+--------------------------+
| usage_statement          |
+--------------------------+
| Ali used ChatGPT-4       |
| Sara used ChatGPT-3.5    |
| John used Claude 3 Opus  |
| Ali used ChatGPT-4       |
| Maria used Gemini Pro    |
| Chen used LLaMA 2        |
+--------------------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

Question 05

```
MariaDB [lab_10_task]> SELECT model_name, LENGTH(model_name) AS name_length, SUBSTRING(model_name, -5) AS last_5_chars
    -> FROM models;
+---------------+-------------+--------------+
| model_name    | name_length | last_5_chars |
+---------------+-------------+--------------+
| ChatGPT-4     |           9 | GPT-4        |
| ChatGPT-3.5   |          11 | T-3.5        |
| Claude 3 Opus |          13 |  Opus        |
| Claude 2      |           8 | ude 2        |
| Gemini Pro    |          10 | i Pro        |
| LLaMA 2       |           7 | aMA 2        |
+---------------+-------------+--------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

QUESTION 06

```
MariaDB [lab_10_task]> SELECT cost, CAST(cost AS CHAR) AS cost_as_string
    -> FROM usage_logs;
+--------+----------------+
| cost   | cost_as_string |
+--------+----------------+
| 0.0200 | 0.0200         |
| 0.0050 | 0.0050         |
| 0.0300 | 0.0300         |
| 0.0150 | 0.0150         |
| 0.0250 | 0.0250         |
| 0.0000 | 0.0000         |
+--------+----------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

QUESTION 07

```
MariaDB [lab_10_task]> SELECT CASE WHEN cost IS NULL OR cost = 0 THEN 'Free Usage' ELSE CAST(cost AS CHAR) END AS cost_display
    -> FROM usage_logs;
+--------------+
| cost_display |
+--------------+
| 0.0200       |
| 0.0050       |
| 0.0300       |
| 0.0150       |
| 0.0250       |
| Free Usage   |
+--------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

QUESTION 08

```
MariaDB [lab_10_task]> SELECT request_time, DATE_FORMAT(request_time, '%d-%b-%Y') AS formatted_date
    -> FROM usage_logs;
+---------------------+----------------+
| request_time        | formatted_date |
+---------------------+----------------+
| 2024-01-10 10:00:00 | 10-Jan-2024    |
| 2024-01-11 12:00:00 | 11-Jan-2024    |
| 2024-01-12 15:30:00 | 12-Jan-2024    |
| 2024-01-15 09:00:00 | 15-Jan-2024    |
| 2024-01-20 18:00:00 | 20-Jan-2024    |
| 2024-01-22 20:00:00 | 22-Jan-2024    |
+---------------------+----------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

QUESTION 09

```
MariaDB [lab_10_task]> SELECT STR_TO_DATE('15-02-2024', '%d-%m-%Y') AS converted_date;
+----------------+
| converted_date |
+----------------+
| 2024-02-15     |
+----------------+
1 row in set (0.001 sec)

MariaDB [lab_10_task]>
MariaDB [lab_10_task]>
```

QUESTION 10

```
MariaDB [lab_10_task]>
MariaDB [lab_10_task]> SELECT model_name, DATE_FORMAT(release_date, '%M %Y') AS release_month_year
    -> FROM models;
+----------------+--------------------+
| model_name     | release_month_year |
+----------------+--------------------+
| ChatGPT-4      | March 2023         |
| ChatGPT-3.5    | November 2022      |
| Claude 3 Opus  | March 2024         |
| Claude 2       | July 2023          |
| Gemini Pro     | February 2024      |
| LLaMA 2        | July 2023          |
+----------------+--------------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

Question 11

```
MariaDB [lab_10_task]> SELECT log_id, DATE_FORMAT(request_time, '%W, %d %M %Y') AS formatted_request_date
    -> FROM usage_logs;
+--------+---------------------------+
| log_id | formatted_request_date    |
+--------+---------------------------+
|      1 | Wednesday, 10 January 2024 |
|      2 | Thursday, 11 January 2024 |
|      3 | Friday, 12 January 2024   |
|      4 | Monday, 15 January 2024   |
|      5 | Saturday, 20 January 2024 |
|      6 | Monday, 22 January 2024   |
+--------+---------------------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

Question 12

```
MariaDB [lab_10_task]> SELECT request_time, DATE_ADD(request_time, INTERVAL 1 WEEK) AS plus_one_week
    -> FROM usage_logs;
+---------------------+---------------------+
| request_time        | plus_one_week       |
+---------------------+---------------------+
| 2024-01-10 10:00:00 | 2024-01-17 10:00:00 |
| 2024-01-11 12:00:00 | 2024-01-18 12:00:00 |
| 2024-01-12 15:30:00 | 2024-01-19 15:30:00 |
| 2024-01-15 09:00:00 | 2024-01-22 09:00:00 |
| 2024-01-20 18:00:00 | 2024-01-27 18:00:00 |
| 2024-01-22 20:00:00 | 2024-01-29 20:00:00 |
+---------------------+---------------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

Question 13

```
MariaDB [lab_10_task]> SELECT request_time, DATE_SUB(request_time, INTERVAL 1 MONTH) AS minus_one_month
    -> FROM usage_logs;
+---------------------+---------------------+
| request_time        | minus_one_month     |
+---------------------+---------------------+
| 2024-01-10 10:00:00 | 2023-12-10 10:00:00 |
| 2024-01-11 12:00:00 | 2023-12-11 12:00:00 |
| 2024-01-12 15:30:00 | 2023-12-12 15:30:00 |
| 2024-01-15 09:00:00 | 2023-12-15 09:00:00 |
| 2024-01-20 18:00:00 | 2023-12-20 18:00:00 |
| 2024-01-22 20:00:00 | 2023-12-22 20:00:00 |
+---------------------+---------------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

Question 14

```
MariaDB [lab_10_task]> SELECT l.log_id, m.model_name, m.release_date, l.request_time, DATEDIFF(l.request_time, m.release_date) AS days_diff
    -> FROM usage_logs l
    -> JOIN models m ON l.model_id = m.model_id;
+--------+---------------+--------------+---------------------+-----------+
| log_id | model_name    | release_date | request_time        | days_diff |
+--------+---------------+--------------+---------------------+-----------+
|      1 | ChatGPT-4     | 2023-03-14   | 2024-01-10 10:00:00 |       302 |
|      2 | ChatGPT-3.5   | 2022-11-30   | 2024-01-11 12:00:00 |       407 |
|      3 | Claude 3 Opus | 2024-03-01   | 2024-01-12 15:30:00 |       -49 |
|      4 | ChatGPT-4     | 2023-03-14   | 2024-01-15 09:00:00 |       307 |
|      5 | Gemini Pro    | 2024-02-01   | 2024-01-20 18:00:00 |       -12 |
|      6 | LLaMA 2       | 2023-07-18   | 2024-01-22 20:00:00 |       188 |
+--------+---------------+--------------+---------------------+-----------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```

Question 15

```
MariaDB [lab_10_task]> SELECT log_id, request_time, response_time, TIMEDIFF(response_time, request_time) AS response_delay, TIMESTAMPDIFF(SECOND, request_ti
me, response_time) AS time_taken_seconds
    -> FROM usage_logs;
+--------+---------------------+---------------------+----------------+--------------------+
| log_id | request_time        | response_time       | response_delay | time_taken_seconds |
+--------+---------------------+---------------------+----------------+--------------------+
|      1 | 2024-01-10 10:00:00 | 2024-01-10 10:00:05 | 00:00:05       |                  5 |
|      2 | 2024-01-11 12:00:00 | 2024-01-11 12:00:02 | 00:00:02       |                  2 |
|      3 | 2024-01-12 15:30:00 | 2024-01-12 15:30:07 | 00:00:07       |                  7 |
|      4 | 2024-01-15 09:00:00 | 2024-01-15 09:00:04 | 00:00:04       |                  4 |
|      5 | 2024-01-20 18:00:00 | 2024-01-20 18:00:06 | 00:00:06       |                  6 |
|      6 | 2024-01-22 20:00:00 | 2024-01-22 20:00:01 | 00:00:01       |                  1 |
+--------+---------------------+---------------------+----------------+--------------------+
6 rows in set (0.001 sec)

MariaDB [lab_10_task]>
```